

RESEARCH AND EDUCATION

Artificial intelligence in the detection and classification of dental caries



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Dental caries is one of the most common infectious chronic diseases, affecting half of the world's population.¹ The accurate and early diagnosis of dental caries is fundamental to determining the most appropriate treatment. Dental caries is commonly diagnosed clinically by visual-tactile assessment and radiographic examinations.² Additional devices that have been used to detect dental caries include digital imaging, light-emitting diode technology, fiber-optic transillumination, and fluorescence cameras and lasers.^{2,3} For proximal caries, when it is hard to inspect the lesion clinically, bitewings radiographs have been used to determine caries depth.^{4,5} Automated diagnostic tools for dental caries have evolved with the advancement of digital technology.⁶

Artificial intelligence (AI) has been defined as “the ability of an engineered system to obtain, understand, and apply facts and abilities learned via knowledge, study, or education that are generally connected with

ABSTRACT

Statement of problem. Automated detection of dental caries could enhance early detection, save clinician time, and enrich treatment decisions. However, a reliable system is lacking.

Purpose. The purpose of this study was to train a deep learning model and to assess its ability to detect and classify dental caries.

Material and methods. Bitewings radiographs with a 1876×1402-pixel resolution were collected, segmented, and anonymized with a radiographic image analysis software program and were identified and classified according to the modified King Abdulaziz University (KAU) classification for dental caries. The method was based on supervised learning algorithms trained on semantic segmentation tasks.

Results. The mean score for the intersection-over-union of the model was 0.55 for proximal carious lesions on a 5-category segmentation assignment and a mean F1 score of 0.535 using 554 training samples.

Conclusions. The study validated the high potential for developing an accurate caries detection model that will expedite caries identification, assess clinician decision-making, and improve the quality of patient care. (J Prosthet Dent 2025;133:1326-1332)

human intellect, and capable of adaptable autonomous acts.”^{4,5} The solutions are not programmed directly into the system. However, with the required data, the AI can decipher those data and manufacture an answer by itself. AI subfields include machine learning (ML), deep learning (DL), neural network, natural language processing, cognitive computing, and computer vision.

ML is the technique that allow computers to learn without being explicitly programmed⁶ and is the foundation of AI algorithms that are trained to detect statistical patterns

Supported by the Ministry of Education and King Abdulaziz University (Saudi Arabia) through institutional fund projects [IFPIP: 1443-165-1442].

Declarations of interest: None applicable for all authors.

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Clinical Implications

Artificial intelligence in the detection and classification of dental caries can increase the diagnostic accuracy and sensitivity to detect caries of dentists. Consequently, this can help with early detection and decision-making for dental caries, reducing biological complications and reducing the need for endodontic treatment.

in a specified set of data.⁷ ML promotes the tactic to a sophisticated level by providing data for a machine to train and adapt fittingly once exposed to new data.^{8,9} The emphasis of ML is mining information from significantly “big data,” to detect and identify underlying patterns by exploiting various measures with applied mathematics to improve the ability to interpret new data and manufacture practical results.^{8,9} ML patterns can be in the form of classification, regression, and/or clustering, either supervised or unsupervised.⁸

DL is a subfield of ML that is inspired by the structure and function of the brain, specifically the neural networks that make up the brain.¹⁰ DL involves training artificial neural networks, which are networks of algorithms that are designed to recognize patterns, on a large dataset. These networks can be used for a range of tasks, such as image recognition, natural language processing, and decision-making. They are called “deep” because they have many layers of processing units, allowing them to learn increasingly complex representations of the data. Neural networks can take selections and return results based on the hidden patterns found within the massive knowledge array.¹⁰

AI can be used in many different clinical dental situations,¹¹ and attempts have been made to detect, classify, and diagnose dental caries via AI, ML, and DL.^{4,5} Most of the published studies regarding AI and the automated detection of dental caries were designed to identify whether dental caries was present or absent.¹²⁻¹⁵ Few studies identified the extension of caries into enamel and dentin.^{12,13} Proximal caries has been evaluated with an AI model by using periapical radiographs and matching the result with a histological examination of the specimens.¹³ The design software program was successful in detecting the depth and extension of 60% of carious lesions in enamel and 97% in dentin from posterior periapical radiographs after segmentation. Likewise, Devito et al¹¹ reported that AI software program displayed better proximal caries diagnosis than a visual examiner using bitewing radiographs.¹²

Few studies have used DL for dental caries detection^{10,14} or progression.¹⁵ Lee et al¹⁴ evaluated the efficacy of deep convolutional neural network (CNN) algorithms for the diagnosis of dental caries from periapical radiographs.¹⁴ The average diagnostic accuracy for

dental caries detection in premolars was approximately 89%, in molars 88%, and together 82%. Sensitivity, specificity, positive predictive value, negative predictive value, receiver operating characteristic curve, and area under the curve were also analyzed for identifying the deep CNN algorithm. Deep CNN algorithms allowed efficient edge detection via multiple convolutional and hidden layers with hierarchical feature pictures. Sensitivity, accuracy, and F measure values for caries detection were 0.84, 0.84, and 0.84, respectively and 0.81, 0.86, and 0.84 for caries segmentation.¹⁰

Distinguishing between enamel and dentin caries is crucial for determining the most appropriate treatment as accurately and early as possible. Detection of dental caries prevents the progression of the disease to the dentin and then to the pulp, with loss of vitality and subsequent endodontic treatment or extraction. AI models achieved a caries diagnosis accuracy between 76% and 88.3%, sensitivity between 73% and 90%, and specificity between 61.5% and 93%.⁷ The classification of dental caries based on the optimal treatment using AI is still undeveloped. Therefore, this study aimed to suggest an automated AI model for caries detection and segmentation based on ML algorithms and to evaluate the effectiveness of the model in identifying and classifying dental caries based on the King Abdulaziz University (KAU)-modified (ICDAS) system from dental bitewings. The null hypothesis was that no difference would be found in the accuracy of caries detection among the four classifications.

MATERIAL AND METHODS

The study was conducted at KAU Dental Hospital in Jeddah, Saudi Arabia. The collected bitewing radiograph datasets were acquired between June 2022 and September 2022 from the Electronic Medical Record System (CS R4 Practice Management Software). All images were collected after permission was granted from the Radiology Department of KAU (IRB approval IFPIP: 1443-165-1442). The dataset consisted of bitewing radiographs, which included images of permanent maxillary and mandibular teeth, with initial, moderate, or advanced carious lesions. The inclusion criterion was bitewing images showing enamel or dentin carious lesions and permanent teeth. The exclusion criteria were periapical radiographs, overlapping images, radiographs with distortions and shadows, and images with indirect restorations. Duplicated images and images where carious lesions were difficult to identify because of technical errors that might have interfered with caries identification were omitted from the dataset to improve accuracy. Image augmentation was applied to improve brightness and contrast for the collected images.

The four main types of labeling tasks are categorization, segmentation, sequencing, and mapping. In the present study, semantic segmentation was used to identify the location, size, and shape of an object in each image by labeling each pixel of an image with a matching label, as in the representative labeled radiographic image (Fig. 1).

After cleaning and augmentation, 2 examiners (A.A.A., W.M.A.) were calibrated before the labeling process. A data annotation was built using a web-based platform (RoboFlow; Roboflow, Inc) that allows users to upload datasets for image annotations, orientations, and size modifications so that it can be used to train an AI model. The 2 calibrated restorative dentists (A.A.A., W.M.A.) manually labeled all images according to the radiographic section of the KAU-ICIDAS (Fig. 1). Five classifications based on the lesion extension were determined. Score 0 (S0), where there was no radiographic evidence of carious lesions; score 1 (S1), where the lesion was within the outer half of the enamel; score 2 (S2), where the lesion was within the inner half of the enamel; score 3 (S3), where the lesion was within the

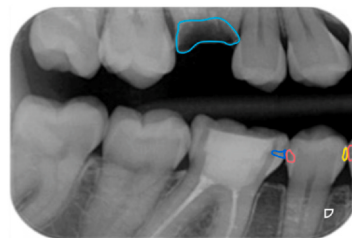


Figure 2. King Abdulaziz University-modified International Caries Detection and Assessment System (KAU-ICIDAS).

outer half of the dentin; and score 4 (S4), where the lesion was within the inner half of the dentin (Fig. 2).

The collected and labeled dataset consists of nearly 554 bitewings. All images were converted from the original format to the Joint Photographic Experts Group (JPEG) file format to unify the quality of the image processing procedure. Furthermore, all images were resized to 512×512. Additionally, their contrast was increased to make it easier for the model to detect initial carious lesions (S1). The dataset is divided into three subsets: train, validation, and test sets.

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Caries Diagnosis



Student's Name:								Computer No.:								Date:							
Patient's File No.:								Gender : M F				Age:				Phone/cell:							

								Occlusal									
								Distal									
								Palatal									
								Mesial									
								Labial/buccal									
18	17	16	15	14	13	12	11		21	22	23	24	25	26	27	28	
48	47	46	45	44	43	42	41		31	32	33	34	35	36	37	38	
								Labial/buccal									
								Mesial									
								Distal									
								Lingual									
								Occlusal									

Caries severity 0: sound 1: enamel lesion visible only after drying 2: enamel lesion visible without drying 3: shadowing of dentin with or without cavitation 4: extensive lesion with visible dentin				Caries activity: A: active I: Inactive				Radiographic examination: 0: no radiographic evidence of lesions 1: lesion within the outer half of enamel 2: lesion within the inner half of enamel 3: lesion within the outer half of dentin 4: lesion within the inner half of dentin				Root caries: 0: sound 1: lesion with discoloration 2: cavitated lesion with 1-2 mm depth 3: cavitated lesion with >2mm depth			
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Recurrent caries: ☐ yes ☐ no **Additional diagnostic aids:** ☐ transillumination ☐ EPT ☐ cold test ☐ heat test ☐ percussion

Criteria	Self-assessment			Instructor's assessment			Feedback	Global Score
	0	1	2	0	1	2		
Severity*								/ 100
Activity*								
Other factors								

Instructor's Name:				Signature & Stamp				Date:			
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Figure 1. Representative image showing King Abdulaziz University-modified International Caries Detection and Assessment System (KAU-ICIDAS) categories of caries classification.

Table 1. Hyperparameters used in individual and ensemble models

Parameter	Model
Epochs	60
Callbacks	Early stopping
Learning rate	0.0001
Optimizer	Adam
Loss	Focal, and Dice loss
Batch size	2
Encoders	ResNet50, ResNext101, Vgg19

The obtained radiographic images were processed using a programming language (Python; Python Software Foundation) in addition to Python-friendly environments (Colab; Google). Additionally, the Keras and Segmentation models library, which had been previously trained on a large general dataset, was used to build the current model. The library used to design the U-Net architecture was segmentation models in Python, which used Keras. The ResNet-50, ResNext101, and Vgg19 networks were all pretrained models in Keras, which were loaded into the segmentation_model module, which provides the U-Net architecture. Hyperparameter variables were set manually with a predetermined value before starting the model training (Table 1). The dataset was divided into 80% for training, 10% for testing, and 10% for validation. All training was

done within a web-integrated development environment for Python (Colab; Google) using a graphics processing unit runtime and the U-Net model (Fig. 3). Three encoders were used (ResNet50, ResNext101, and Vgg19), and all were pretrained on ImageNet weights. The three U-Net models were then joined to get the final results. As the training method is highly dependent on the size and quality of the data, the transfer learning process was performed to improve accuracy given the limited number of labeled images.

To evaluate the model performance, the data of the test set were statistically analyzed, including for sensitivity, precision, true-positive (TP), false-positive (FP), false-negative (FN), F1 score, and Intersection over Union (IOU). Each radiographic image was compared with the base labeling provided previously. The testing dataset was not observed by the model during the training phase to enable the model to generalize to all future data. Recall was defined as a metric that quantified the number of correct positive predictions made from all positive predictions that could have been made: (Recall [sensitivity]=TP/TP+FN). Precision was defined as a metric that quantifies the number of correct positive predictions made: (Precision=TP/TP+FP).

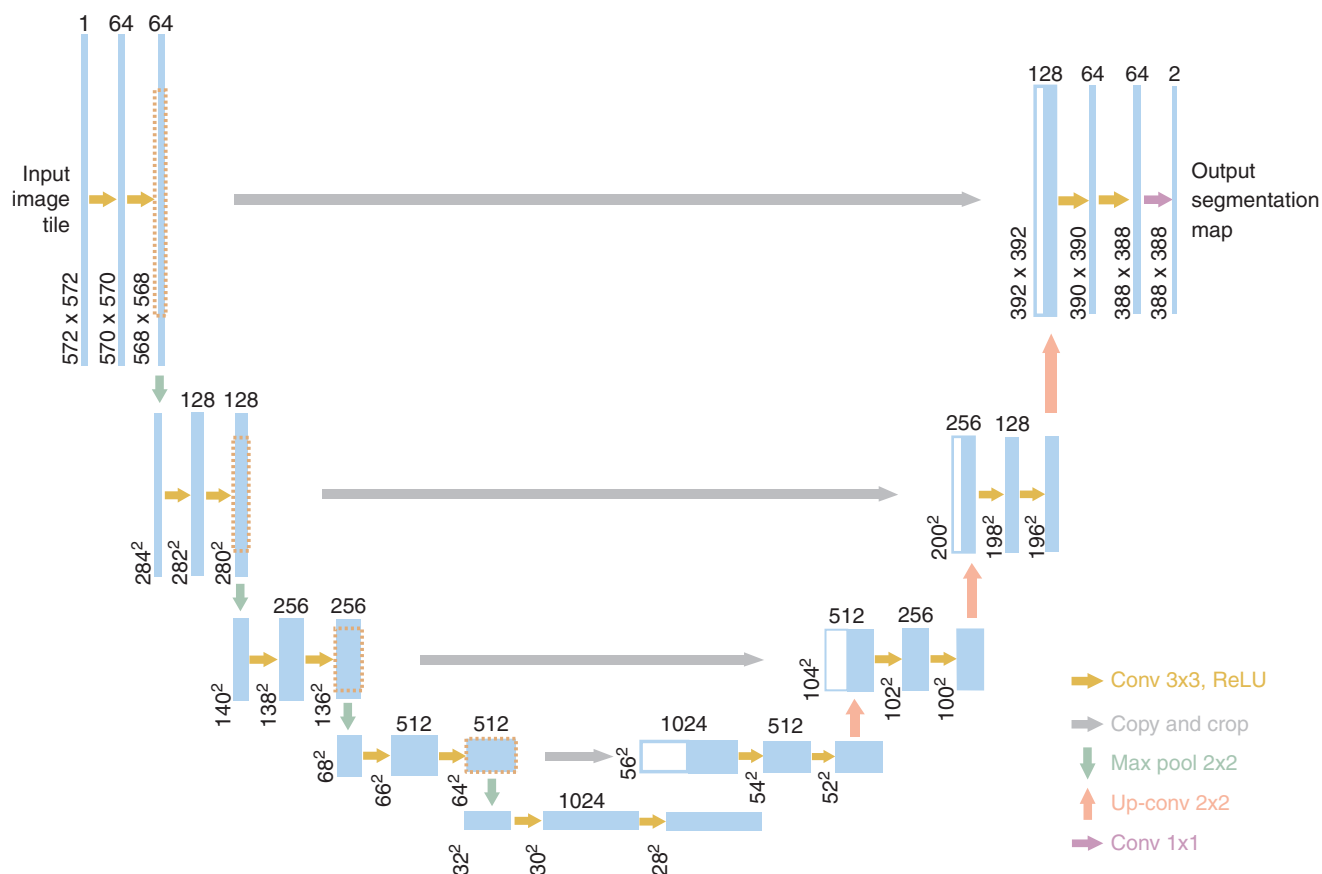


Figure 3. U-Net Architecture. Three U-Net models assembled to obtain final results. ResNet50, ResNext101, and Vgg19 used as encoders, all pretrained on ImageNet weights. Ensemble learning combines several weak models to produce model with stronger predictive power.

The F1 score was expressed as a metric that combines recall and precision into a solitary score by calculating the harmonic mean of precision and recall. It was used instead of accuracy because of the imbalance observed in the datasets: (F1 score = $[2 \times \text{Precision} \times \text{Recall}] / [\text{Precision} + \text{Recall}]$).

The IoU score, an extensively used metric that measured the likeness between the predicted area and the ground-truth area in which the intersection is divided by the union of the two areas, was leveraged as the primary evaluation metric. IoU can be rephrased in terms of true/false positives/negatives as $\text{IOU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$.

RESULTS

The supervised learning algorithm model was created and trained on the labeled data set. The results were evaluated and completed on the test set using the IoU score and F1 score. The IoU averaged 96% for no caries (S0), 21% for S1 (Outer Enamel), 23% for S2 (Inner Enamel), 35% for S3 (Outer Dentin), and 41% for S4 (Inner Dentin). The means for IoU and F1 scores for each model separately are shown in Table 2, and a summary of the statistical results for the overall model is reported in Table 3. The model was very efficient in recognizing whether a shadow represented a carious lesion or not. Moreover, the model was able to accurately detect carious lesions more with a S3 (35%) and a S4 (41%) (Fig. 4). Misclassification occurred mostly between S1 and S2 carious lesions (Fig. 5). As a standalone model, the model with resnext101 as its encoder had the best performance on the datasets.

DISCUSSION

Recently, AI has substantially expanded in dentistry in terms of the assessment, localization, classification, diagnosis, and estimation of disease progression.¹⁶ The

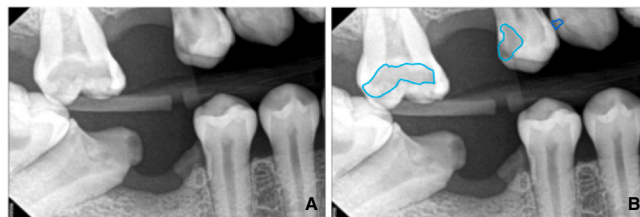


Figure 4. Results of models. A, Testing image. B, Testing label.



Figure 5. Misclassification of models. A, Testing image. B, Testing label.

authors are unaware of a previous study using the KAU-ICIDAS to apply, train, and test the software program to detect and classify proximal dental caries from bitewing radiographs. The accuracy of caries detection was an IOU mean score of 0.55 for proximal carious lesions on a 4-class segmentation task and a mean F1 score of 0.535. Therefore, the null hypothesis was rejected.

ML is a comprehensive term that incorporates a variety of algorithms and techniques designed to enable computers to learn from data and make predictions or decisions without explicit programming. In the context of caries detection, ML algorithms can analyze various data types, such as clinical and radiographic findings, to generate predictive models for caries risk assessment. ML models can identify patterns and correlations between various features and caries occurrence, thereby facilitating the prediction of caries development or the identification of individuals at a higher risk of caries.

DL is a subset of ML that simulates the neural structure and function of the human brain using artificial

Table 2. Mean of IOU and F1 score for each model

Encoder	Mean F1	Mean IOU	Mean IOU for No Caries	Mean IOU for Score 4	Mean IOU for Score 3	Mean IOU for Score 2	Mean IOU for Score 1
Resnet50	0.457	0.5455288	0.961	0.35	0.22	0.26	0.24
ResNext101	0.612	0.5660605	0.952	0.47	0.30	0.21	0.19
Inception-ResNet-v2	0.538	0.4538291	0.96	0.38	0.25	0.22	0.21

IOU, intersection-over-union.

Table 3. Mean of IOU and F1 score for overall model

Mean F1	Mean IOU	Mean IOU for No Caries	Mean IOU for Score 4	Mean IOU for Score 3	Mean IOU for Score 2	Mean IOU for Score 1
0.535	0.551	0.957	0.41	0.35	0.23	0.21

IOU, intersection-over-union.

neural networks. DL algorithms are designed to process and extract information from high-dimensional and complex datasets. DL algorithms can analyze large datasets of dental images, such as intraoral radiographs and dental photographs, to accurately identify and classify carious lesions. DL models excel at autonomously learning and extracting intricate features from input data, enabling a more thorough evaluation of carious lesions, including their size, location, and severity.

In comparison with the present study, Casalegno et al¹⁶ used a DL model to train 217 data sets to detect dental caries using near-infrared light transillumination (NILT) images. On a 5-grade segmentation test, a total average IOU rate of 72.7% was obtained, with an IOU rate of 49.5% for interproximal and 49.0% for occlusal caries.¹⁷ For the categorization of the presence or absence of dental caries, their model achieved an area under the ROC rate of 83.6% occlusally and 85.6% interproximally.

Likewise, Schwendicke et al¹⁷ used the DL-CNN method using 2 encoders, Resnet 18 and Resnet 50, and reported a similar area under the ROC curve performance of 0.74 at 95% CI.¹⁸ Caries detection was performed in 226 data sets of NILT images of 113 extracted premolars and 113 molar teeth. Sensitivity was reported to be 0.59 and specificity 0.76.

The DL model has been applied to segment tooth caries on dental bitewing radiographs and compare the mean accuracy value between DL and the visual detection of dental caries.¹⁸ The U-Net model was used to train 3293 annotated bitewing radiographs. The accuracy was 0.80, sensitivity was 0.75, specificity was 0.83, and the F1 score of the AI model was found to be 0.73. The mean accuracy value of dentists was significantly lower at 0.71.

An AI model was created to detect caries using VGG-16 and segment them using the U-Net model.¹⁰ To develop both models, 10 data augmentation methods trained 613 datasets. In the internal dataset, the detection sensitivity, precision, and F1 score of the model were 0.84, 0.84, and 0.84, respectively, whereas the segmentation of the model sensitivity, precision, and F1 score were 0.81, 0.86, and 0.84.¹⁰ Five distinct observers with diverse experiences were used to compare AI models. In the external dataset, the detection model's sensitivity, precision, and F1 score were 0.77, 0.78, and 0.78, respectively, while the segmentation model's sensitivity, precision, and F1 score were 0.76, 0.87, and 0.81. When their F1 scores were compared, the AI models outperformed an assistant professor of dentomaxillofacial radiology with 2 years of experience and an assistant professor of restorative dentistry with 3 years of experience. Limitations of the present model included that overexposed or underexposed regions in

the radiographs tended to be determined as physically unrealistic labeling artifacts, as did overlapping proximal surfaces. Furthermore, the limited size and information of the lowest truth labels of the dataset was a limitation. Therefore, the collection of numerous new training samples would be needed to improve the performance of the model in detecting dental caries as early as possible.

Tooth overlap in bitewing radiographs can significantly affect the precision and trueness of AI caries detection. Overlapping can obscure areas of dental caries, making it difficult for an AI algorithm to determine the presence and severity of caries accurately. When teeth overlap, shadowing or overlapping pixels can occur on the radiograph, making it difficult for the AI algorithm to differentiate between healthy and carious tooth structure. Moreover, overlapping teeth can also result in false positives, where the algorithm may detect cavities in areas where none exist because the overlapping pixels create the illusion of decay. To address these issues, AI caries detection algorithms may use image segmentation and feature extraction to differentiate between healthy and carious tooth structure in overlapping regions. These methods may require considerable training data and processing power to effectively detect and classify caries in overlapping regions. AI caries detection may enhance dental diagnosis; however, the inability of bitewing radiographs to capture overlapping teeth might make identification of caries difficult.¹³

ML algorithms concentrate on recognizing patterns and making predictions based on labeled data. In contrast, DL algorithms, a subset of ML, excel at autonomous learning and extracting complex features from unlabeled data. DL algorithms are especially efficient at analyzing large datasets of dental images, providing accurate caries detection and the comprehensive characterization of carious lesions. ML incorporates a broader range of algorithms and techniques, including DL, and can be applied to different categories of data to predict caries risk and make informed decisions based on various clinical and demographic factors.

AI and visual examinations detect the severity of caries in distinct ways. AI algorithms are trained on massive datasets and are capable of swiftly analyzing vast amounts of data, which may allow them to detect caries with a high degree of accuracy. AI algorithms nevertheless rely on the integrity of the training data and the algorithms themselves. AI caries detection systems facilitate detecting dental caries at an earlier stage. Moreover, AI can quickly and objectively analyze images in difficult areas where the human observer may have difficulty detecting them. Visual examinations may accurately detect caries severity using visual inspection, an explorer, dental radiographs, patient history, and risk

factors. However, visual examinations can be subjective and variable, as clinicians with varying levels of experience may interpret the presence and severity of dental caries differently. Overall, both AI caries detection and human visual examinations can be used in conjunction to improve the precision and efficacy of dental diagnosis and treatment planning. The model should be trained further by adding images on the proximal surfaces or using dynamic video feedback instead of static images. This will enable simultaneous AR prediction.

As the use of AI in caries detection continues to evolve, more research is required to validate its accuracy relative to traditional methods, to investigate the use of AI in a variety of populations and contexts, and to integrate AI with current clinical practice guidelines. Future recommendations should also include guidelines for the responsible use of AI in dentistry, including the protection of patient data and the provision of informed consent. Strategies for reducing cost and guaranteeing the availability of AI technology should be developed.

CONCLUSIONS

Based on the findings of this study, the following conclusions were drawn:

1. Computer-assisted tools for detecting caries from the semantic segmentation of bitewing radiographs based on a DL approach should facilitate the diagnosis and classification of dental caries.
2. The tools should help in appropriate treatment planning and in monitoring the progression of the disease.
3. The IOU averaged 96% for no caries (S0), 21% for score 1 (Outer Enamel), 23% for score 2 (Inner Enamel), 35% for score 3 (Outer Dentin), and 41% for score 4 (Inner Dentin).

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CRediT authorship contribution statement

All authors gave their final approval and agree to be accountable for all aspects of the work.

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<https://doi.org/10.1016/j.prosdent.2023.07.013>